

Data Analysis with Python

Week 1: Intro to Programming

Justinas Grigaitis, 2026-09-17

Word Guesser

From idea to an app in 15 min

Word Guesser

What have we learnt?

- Algorithmic thinking
- Objects, data structures
- AI-assisted coding
- UI/UX
- Version control
- Deployment

Word Guesser

Poor guy in 18th century

Word



j a _ _

Wrong guesses: 5 / 6

Guessed letters: a b c d j o r

Nice! 'j' is in the word.

Guess

Keyboard

Q

W

E

R

T

Y

U

I

O

P

A

S

D

F

G

H

J

K

L

Z

X

C

V

B

N

M

<https://hangman-game-test.streamlit.app/>

Data Analysis with Python

What is data? What is data analysis?

“Data is a collection of facts, numbers, words, observations or *other useful information*. Through **data processing** and **data analysis**, organizations transform raw data points into **valuable insights** that **improve** decision-making and drive better business outcomes.” (emphasis mine)

<https://www.ibm.com/think/topics/data>

•

Why programming (in 2026)?

Why programming?

Universal character builder

- Algorithmic thinking
- Attention to details
- Efficiency
- Patience
- Creativity
- In 2026...
 - Productivity explosion
 - Language of AI

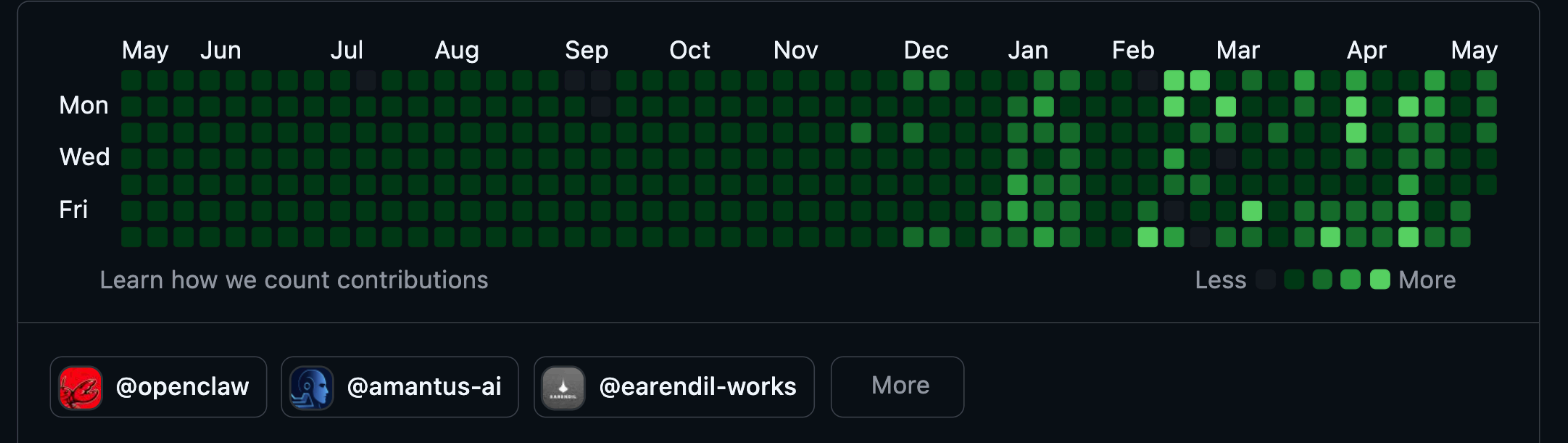


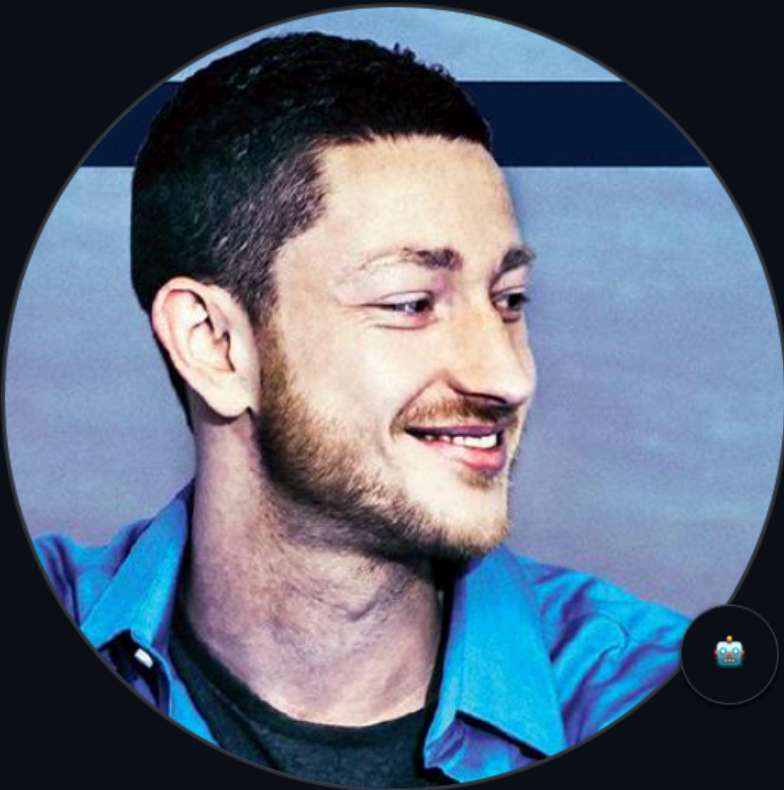
Why programming?

High demand skill

- Software isn't going away, it grows exponentially
- Getting a job **without** programming fundamentals will be harder and harder
- Hierarchy: *everyone → data analyst → data scientist → software engineer*
 - New technologies change the nature of each position
 - Programming skills are essential in levelling up
- Progressive steps - understanding and mastery is what matters at any level!

290,205 contributions in the last year





Peter Steinberger

steipete · he/they

Follow

Sponsor

Came back from retirement to mess with AI. Clawdfather @openclaw Previously: Founder of @PSPDFKit.

49.8k followers · 109 following

steipete / README.md

Hi, I'm Peter 🙋

📍 Vienna ↔ London | 🧠 Polyagentmorous builder | 🚀 Ex-PSPDFKit Founder

Now at OpenAI, working on agents; stewarding OpenClaw as open and independent.

SwiftTypeScriptJavaScriptNode.jsCodexCLImacOSSwiftUIWeb

Deep in vibe-coding mode – building AI-powered developer tools at ludicrous speed. After 13+ years shipping native iOS, modern web feels like a breath of fresh air.

🔍 [sweetistics.com](#) (closed source) – AI-powered Twitter platform with analytics/ops stack.

Start Here

🦖 OpenClaw (370k+ stars) - the AI that actually does things

🗄️ CodexBar (11.9k+ stars) - keep agent limits in view

🦇 ClawHub (8.5k+ stars) - OpenClaw skill directory

🌀 gogcli (7.2k+ stars) - Google in your terminal

👉 summarize (5.8k+ stars) - point at any URL or file, get the gist

📖 agent-rules (5.6k+ stars) - shared rules/knowledge for coding with agents

🚀 Peekaboo (3.4k+ stars) - macOS screenshots & GUI automation for agents

Current Projects

- 🦖 [OpenClaw](#) - the AI that actually does things
- 🦇 [ClawHub](#) - OpenClaw skill directory
- 👉 [ClawSweeper](#) - Weekly issue and PR triage bot
- 📦 [Crabbox](#) - Warm a box, sync the diff, run the suite.
- 🌿 [ask-molty](#) - Ask Molty, shipped as an OpenClaw workspace
- 🖥️ [VibeTunnel](#) - Turn any browser into your terminal; command agents from the road (vt.sh)
- 🗄️ [CodexBar](#) - May your tokens never run out—keep agent limits in view.
- 🚀 [Peekaboo](#) - Lightning-fast macOS screenshots & GUI automation (MCP + CLI)
- 👉 [summarize](#) - Point at any URL or file. Get the gist.
- 🗄️ [RepoBar](#) - CI, PRs, releases—at a glance
- 📖 [acpx](#) - Headless CLI client for stateful ACP sessions
- 🔒 [fs-safe](#) - Race-resistant root-bounded filesystem primitives for Node.js
- 📦 [gitcrawl](#) - Local-first GitHub issue and pull request crawler for maintainer triage
- 🌀 [gogcli](#) - Google in your terminal (gog) (Gmail, Calendar, Drive, Contacts, Tasks, Sheets, Docs, Slides, People)
- 👻 [Poltergeist](#) - The ghost that keeps your builds fresh—universal hot reload & file watcher
- 📱 [wacli](#) - WhatsApp CLI: sync, search, send
- 📖 [wacrawl](#) - Read-only WhatsApp Desktop archive extractor and encrypted backup
- 🐦 [birdclaw](#) - Twitter archive archaeology with Git-backed memory
- 🦊 [discrawl](#) - Mirror Discord into SQLite; search server history locally
- 📖 [slacrawl](#) - Slack history, clawed into SQLite
- 📖 [notcrawl](#) - Local-first Notion crawler into SQLite and Markdown
- 🗣️ [sag](#) - ElevenLabs speech with mac-style say UX; streams to speakers by default
- 🗣️ [Brabble](#) - Wake-word voice daemon for macOS; transcribes locally and fires configurable hooks
- 📞 [vox](#) - Let your agent make the phone call
- 🎧 [sonoscli](#) - Control Sonos speakers: discover, group, queue, play Spotify
- 👉 [ElevenLabsKit](#) - ElevenLabs voices on tap—SwiftPM-friendly, streaming-native.
- 🎧 [songsee](#) - Spectrograms so pretty your ears get jealous
- 📍 [goplaces](#) - Google Places API (New) client + CLI
- 📖 [bslog](#) - Better Stack logs without the browser spelunking
- 🌀 [gifgrep](#) - GIF search for terminals: CLI output + TUI with inline previews
- 📱 [camsnap](#) - RTSP snapshots, clips, motion CLI (Tapo-friendly)
- 🎧 [spogo](#) - Spotify, but make it terminal
- 👉 [ordercli](#) - Your takeout timeline, in the terminal
- 🎧 [blucli](#) - Play, group, and automate BluOS
- 🤖 [macOS Automator MCP](#) - Your Friendly Neighborhood RoboScripter™
- 👉 [Claude Code MCP](#) - One-shot MCP server for Claude Code (an agent inside your agent)
- 👉 [AXorcist](#) - The power of Swift compels your UI to obey!
- 🎧 [Tachikoma](#) - Modern Swift AI SDK
- 📖 [tokentally](#) - One tiny lib for LLM token + cost math
- 👉 [osc-progress](#) - Tiny lib for OSC 9;4 terminal progress.
- ✂️ [Trimmy](#) - “Paste once, run once” — flattens multi-line shell snippets so they execute
- 🖥️ [TauTUI](#) - Swift-native TUI that won't tear
- 🗄️ [Commander](#) - Swift-first parsing, zero forks
- 📖 [remindctl](#) - Apple Reminders from the terminal
- 🗄️ [mcporter](#) - Call MCPs from TypeScript or package them as a CLI
- 📖 [canvas](#) - A browser canvas where agents can draw conclusions
- 🍪 [Sweet Cookie](#) - Inline-first browser cookie extraction—no native addons
- 🍪 [sweetcookie](#) - Browser cookies, baked fresh in Go
- 🍪 [SweetCookieKit](#) - Native macOS cookie extraction for Safari, Chromium, and Firefox
- 🔍 [sweetlink](#) - Playwright vibes in your current tab; close the agent loop
- 🗣️ [bird](#) - Fast X CLI for tweeting, replying, and reading (had to make it private)
- 🗣️ [oracle](#) - Whispering your tokens to the silicon sage
- 👉 [tmuxwatch](#) - Lightweight TUI to watch tmux sessions
- 📖 [agent-scripts](#) - The toolbox my agents reach for first
- 📖 [agent-rules](#) - Shared rules/knowledge for coding with agents
- 🎧 [Markdansi](#) - Wraps, colors, links—no baggage.

Why programming in the age of AI?

High demand skill

- ~~Hierarchy: everyone → data analyst → data scientist → software engineer~~
 - AI has surpassed humans in this technical skill hierarchy
 - AI enables access and raises the bar
 - AI requires higher order skills (levels of abstraction)
- What remains fundamental human?
 - **Responsibility:** *Does it make sense? Is it accurate?*
 - **Direction:** *What's the point? Why do this at all?*

What are people in tech saying?

Boris Cherny (Anthropic) interview in May 2026:

“[Building software] will become a skill like sending a text message [...]. The best person to write accounting software is not an engineer, it’s a really good accountant [...]. **coding is the easy part, knowing the domain is the hard part.**”

Good news

It's never too late to learn coding!

How and what to learn?

Fundamentals, infrastructure, vocabulary

- Learning to code has never been more fun (still Type 2 though)
- Fundamentals more important than ever
 - Transferrable concepts across many programming languages
- Scope of projects is unlimited from Day 1
 - Planning, choosing the right tools, building infra, and version control
- Purpose, creative ideas, and consistency
 - Easy to get overwhelmed or discouraged by AI

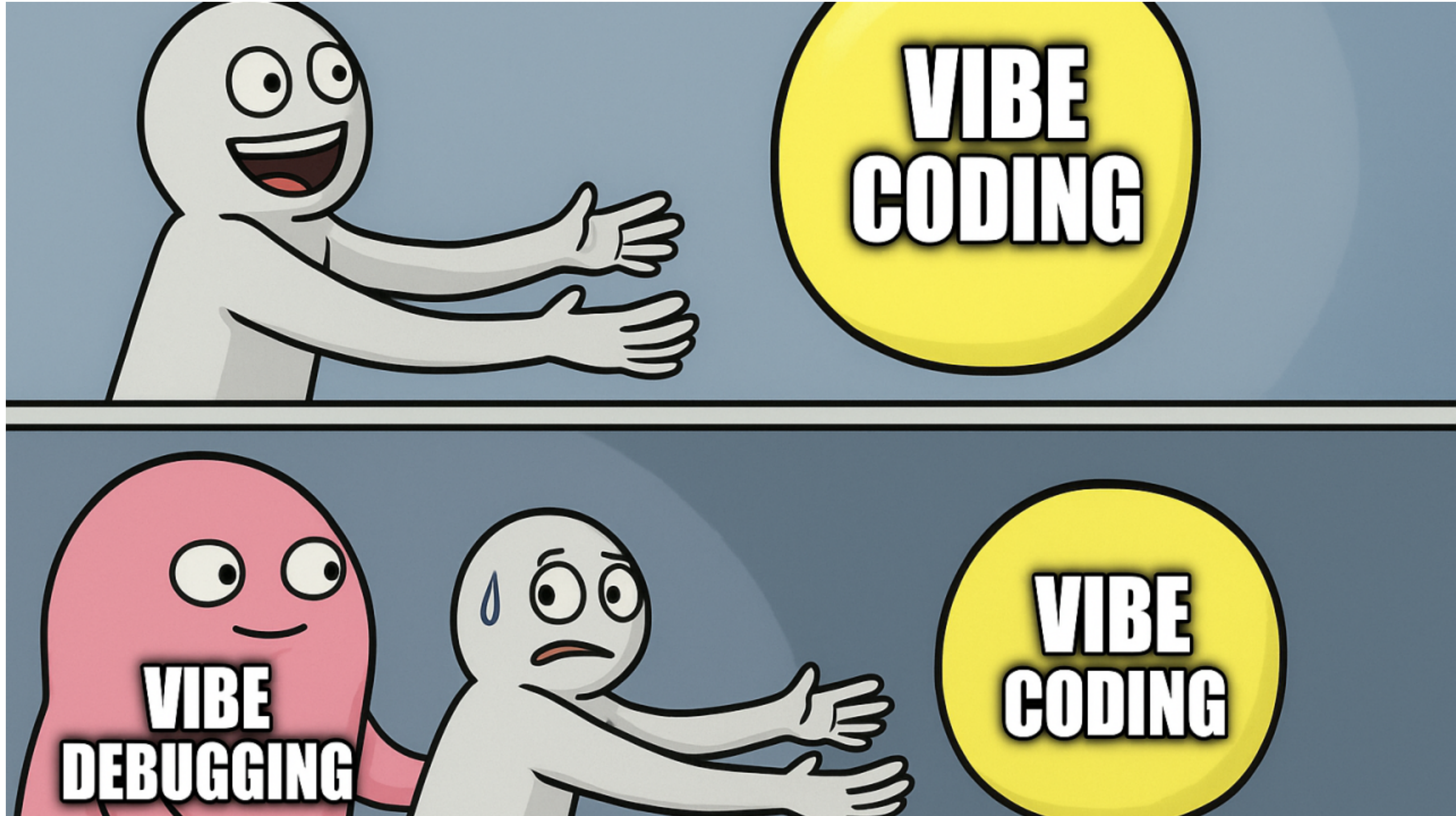
“Vibe coding” is the future?

Software is Changing (Again)

https://youtu.be/LCEmiRjPEtQ?si=F_1RM1qALxw4J52v

- “Vibe coding” = coding without understanding code (Lovable, Replit, Claude Caude)
- “Vibe coding” $\neq$ AI-assisted programming (“agentic engineering”)
- “Vibe coding” is easy -> everyone can do it -> cheap / useless skill
 - Convince why you can do it better than anyone else
 - Figure out what separates you from these 9-13 olds ([link](#))
- “Old school” coding also brings a lot of satisfaction!
- Nice blog post ([link](#))

Believe that effort matters!



Introductions

Getting to know each other

My background

Anyone can learn programming!

- Education and experience
 - Master's + PhD in Economics (UZH, 2022-present)
 - NERA Economic Consulting (NYC, 2019-2022)
 - Sciences Po - Columbia University Dual BA (2014-2019)
- Programming journey
 - Intro to Python at Columbia → data science internship → lab experiment for BA thesis
 - New employees bootcamp at NERA (2020-2022)
 - Interactive learning platforms at UZH (since 2024)
 - Data Analysis for Python (since 2025)

How are you doing?

Heart heart I miss you all

And.....

"I finally started learning something useful"

Here I am this is me



And here are your useful links for today

Recent projects

Research Software Engineering

<https://praxai.ch/>

- Columbia personal page
- PhD Econometrics
- *Empirical Methods Platform*
- *Makro2 Platform*
- India project
 - Cognitive skill measures
 - Math chatbot
- *PraxAI*

**Empirical Methods**
Prof. Gregory Crawford

 Lecture Mon · 10:15–11:45 ·  KOL-F-121 (live) / KOL-H-312 (live video)

 TA sessions Wed · 14:00–15:45 ·  SOD-1-102 and Thu · 16:15–18:00 ·  KOL-F-118


Lectures


Lecture Recap


Weekly Check-in


AI Assistant


Quiz


Homework


Learning Analytics

Professor Ulrich Woitek

TA Gustav Pirich

Speed Learning

5 x 2 min: Hobbies | Research | Professional

Logistics

Not your regular Econ course

Learning objectives

How to think like a programmer

- Have fun, “fall in love” with coding, and do cool projects!

Master Python **fundamentals** and **algorithmic** thinking

Familiarise with key **concepts** and **tools** for **data analysis**

Create a **meaningful** final **project**

- Instructors/peers: inspiration, guidance, and accountability.
- Choose what works for you (goals and style)!

Course outline

From zero to hero

- Weekly meetings Thu 14:00-16:00 at SOF-G-21
 - Weeks 1-2: programming fundamentals
 - Weeks 3-4: data analysis
 - Weeks 5-6: data visualization and projects
- Final project (groups of 2-3, exact requirements TBA)
 - Presentations on Dec 4 (12:00-16:00) at KOL-H-309
 - Report due on Dec 11 at 12:00 (noon) on OLAT

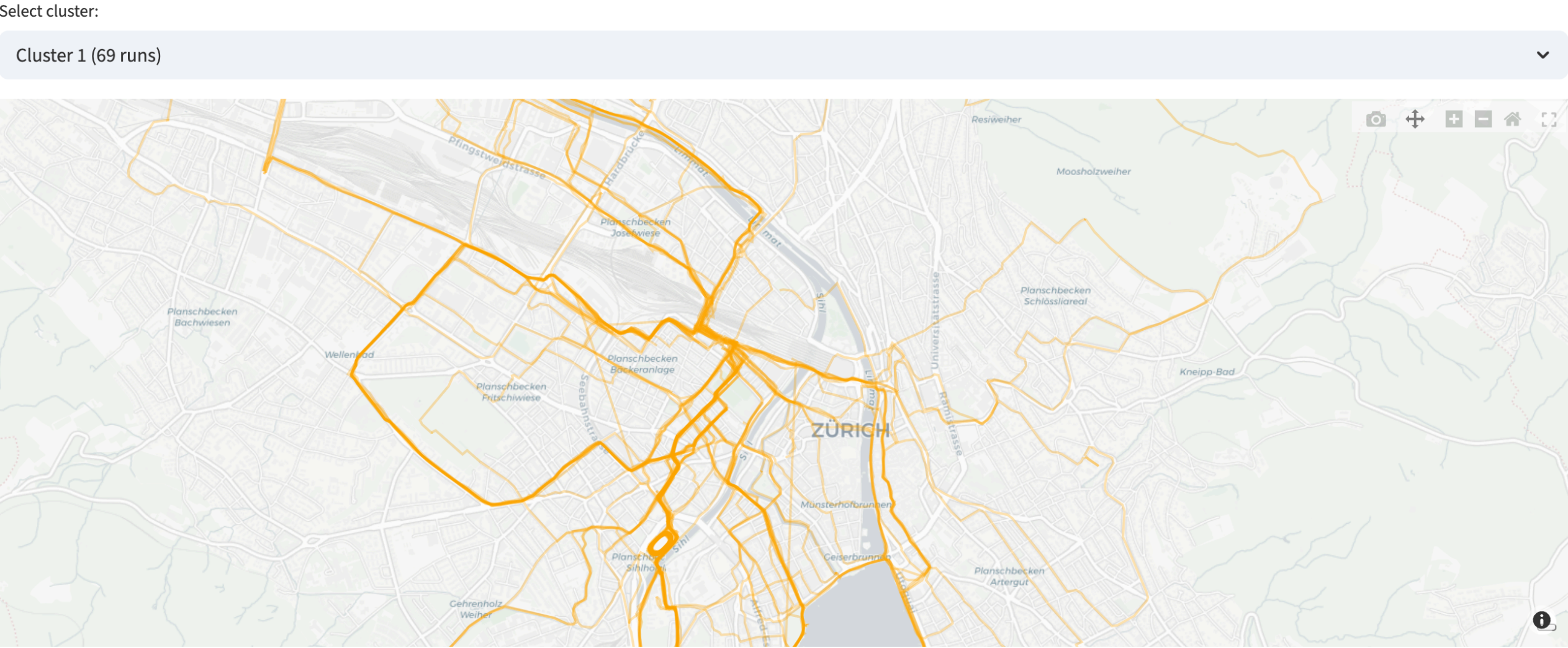
Skill assessment

“A” for effort (and competition :))

- 30% weekly progress (5% x 6 weeks)
 - Homework, self-assessment, project pitches
- 20% fundamentals quiz (October 15)
- 50% final project (December 4 and 11)
 - Web application: hobby-, research- or business-oriented
 - Define your own goals and AI use in advance
 - Continuous effort is what matters



Student projects



Cluster Statistics

Avg Distance (Cluster) 7.9 km	Avg Pace (Cluster) 5:38 min/km	Fastest Pace 4:21 min/km 6.4 km
Avg Elevation Gain 48 m	Median Start Time 16:03	Most Frequent Weekday Tuesday

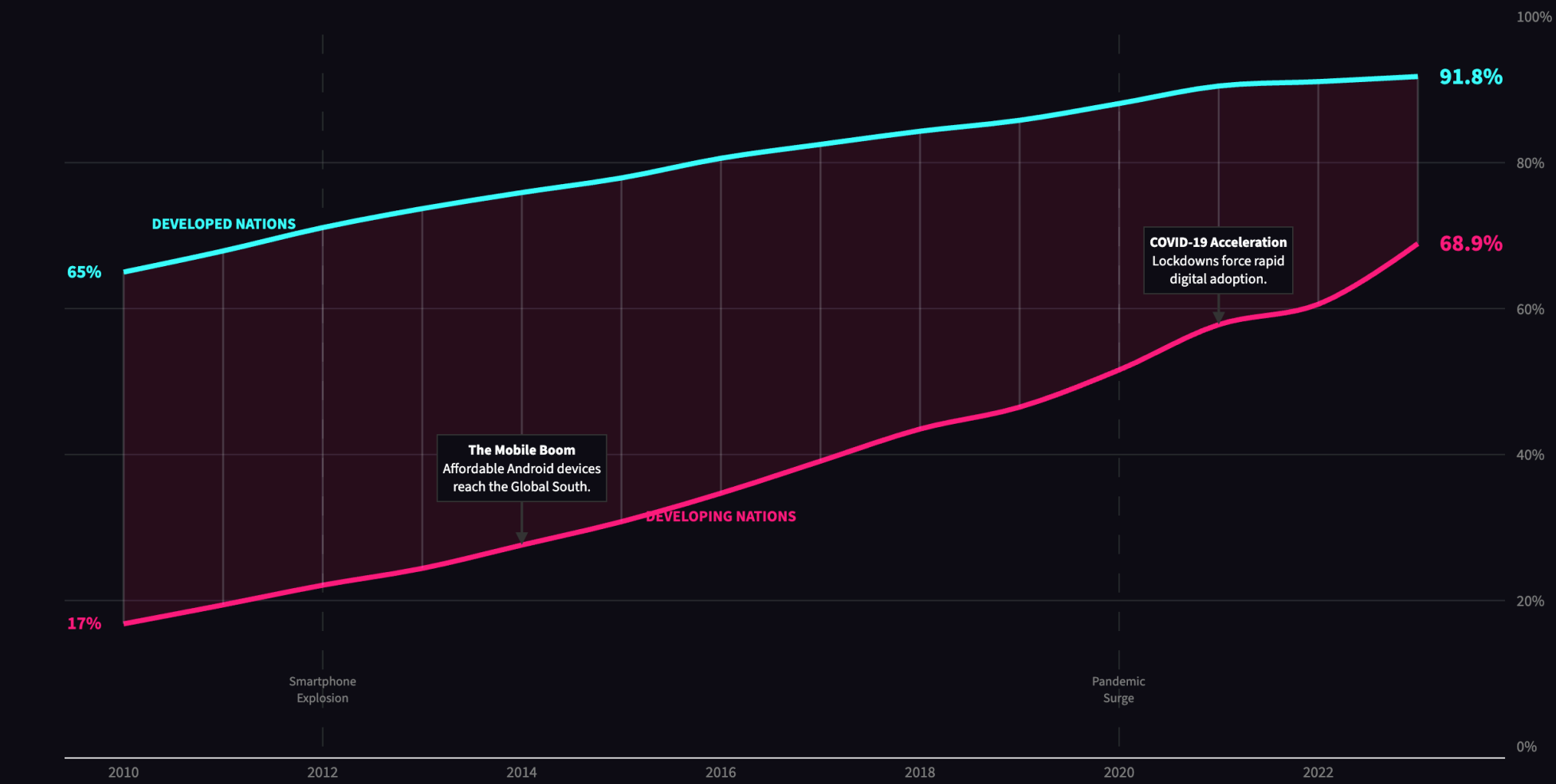
<https://runningthenumbers.streamlit.app/>

Question 3: The Digital Divide

Is the gap between developed and developing countries closing?

The Great Convergence

Global internet adoption rates are harmonizing, though equality remains distant.



Yes, the gap is closing. In 2010, the digital divide was a staggering 48.2 points. Today, **developing nations** have surged to 68.9% connectivity, shrinking the gap to just 22.9 points. While **developed nations** approach saturation, the rest of the world is catching up at 7.5x the speed.

<https://github.com/kevindrmo/kldr-project>

Student projects

Key Figures December 2025

① Percentage change (Δ) is calculated as month-to-month change compared to the previous month.

Total Patient December 2025

210

↓ -5.41%

Avg.Treat.Cost/Patient December 2025

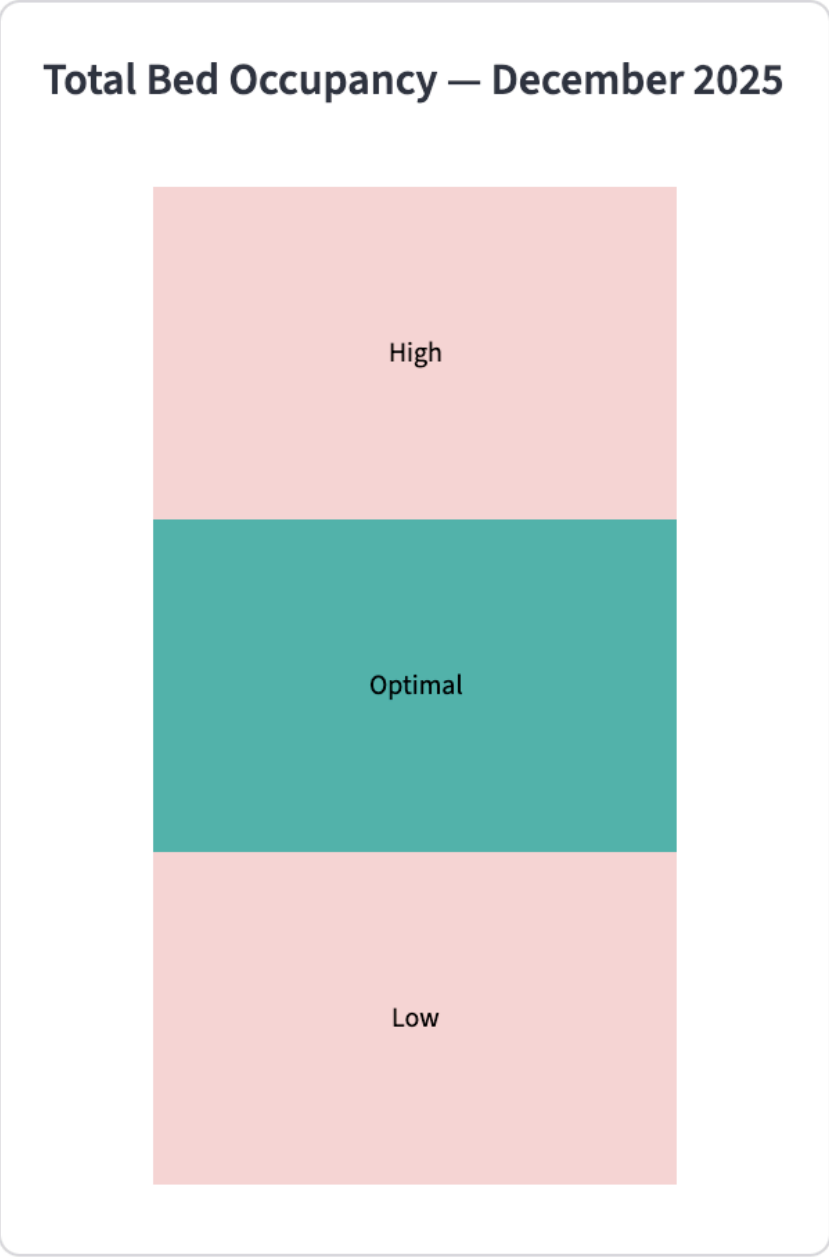
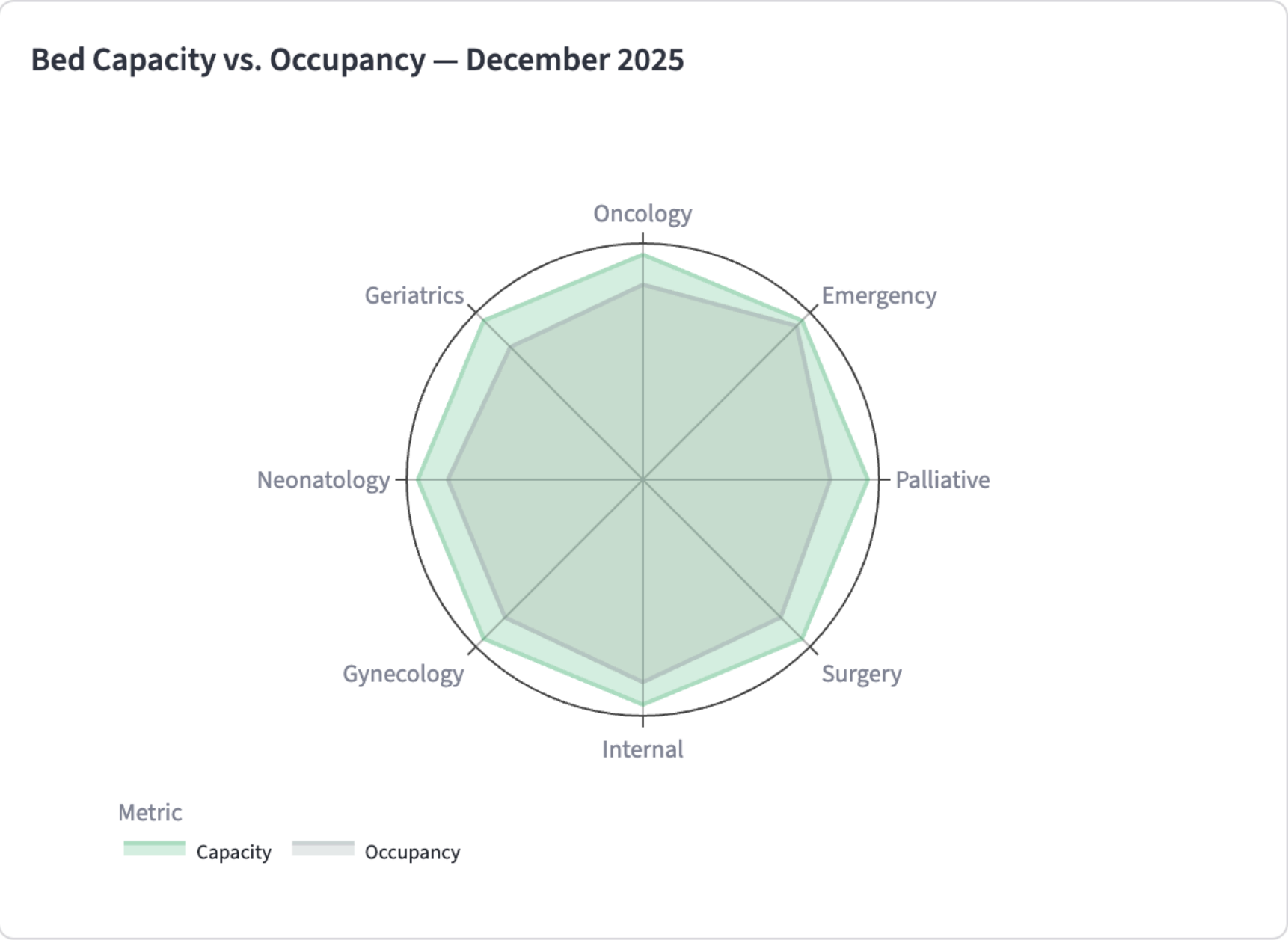
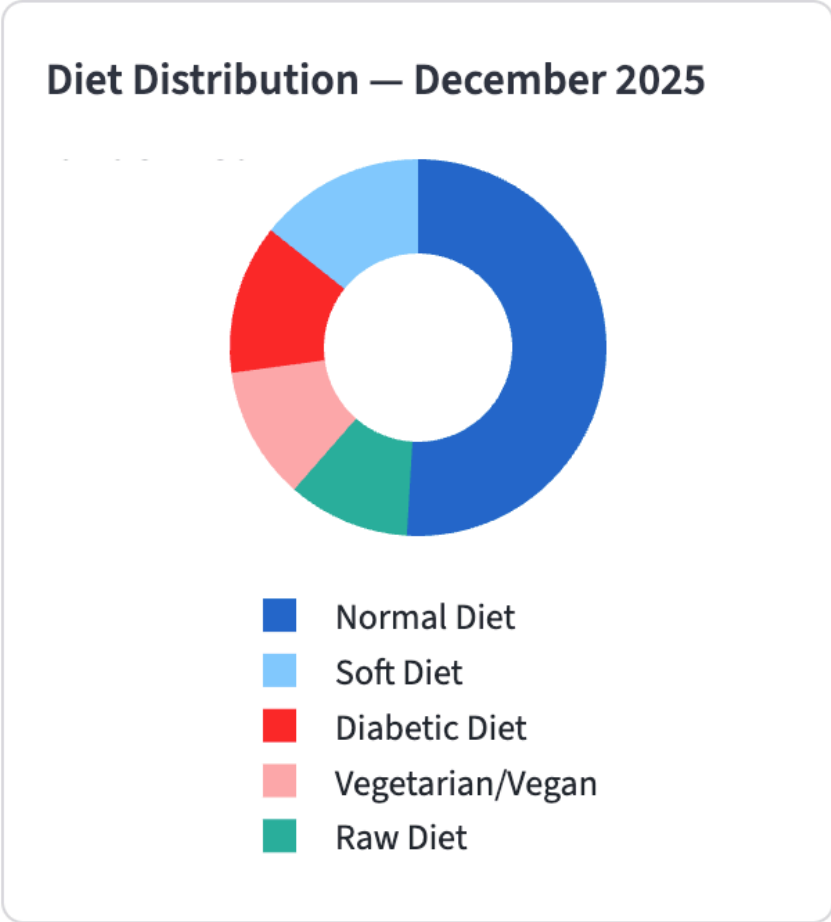
109,766.68

↑ 7.11%

Total Staff December 2025

694

↑ 0.14%



⚠ Critical occupancy in: Emergency

Resources

Where to look for help?

- Textbook: How to Think Like a Computer Scientist
 - O'Reilly books (eg1, eg2)
- More exercises: W3 Schools Python
- Pandas: User Guide, community tutorials
- AI (chat / agents)
- Humans! (Instructors/peers)

Learning to
code is easy



Anyone can
learn how!



Just go to
my website



...and buy
my course



@santamariacode

ProgrammerHumor.io

Excited?

Let's chat!

Short Break

Intro to Python afterwards

Intro to Programming

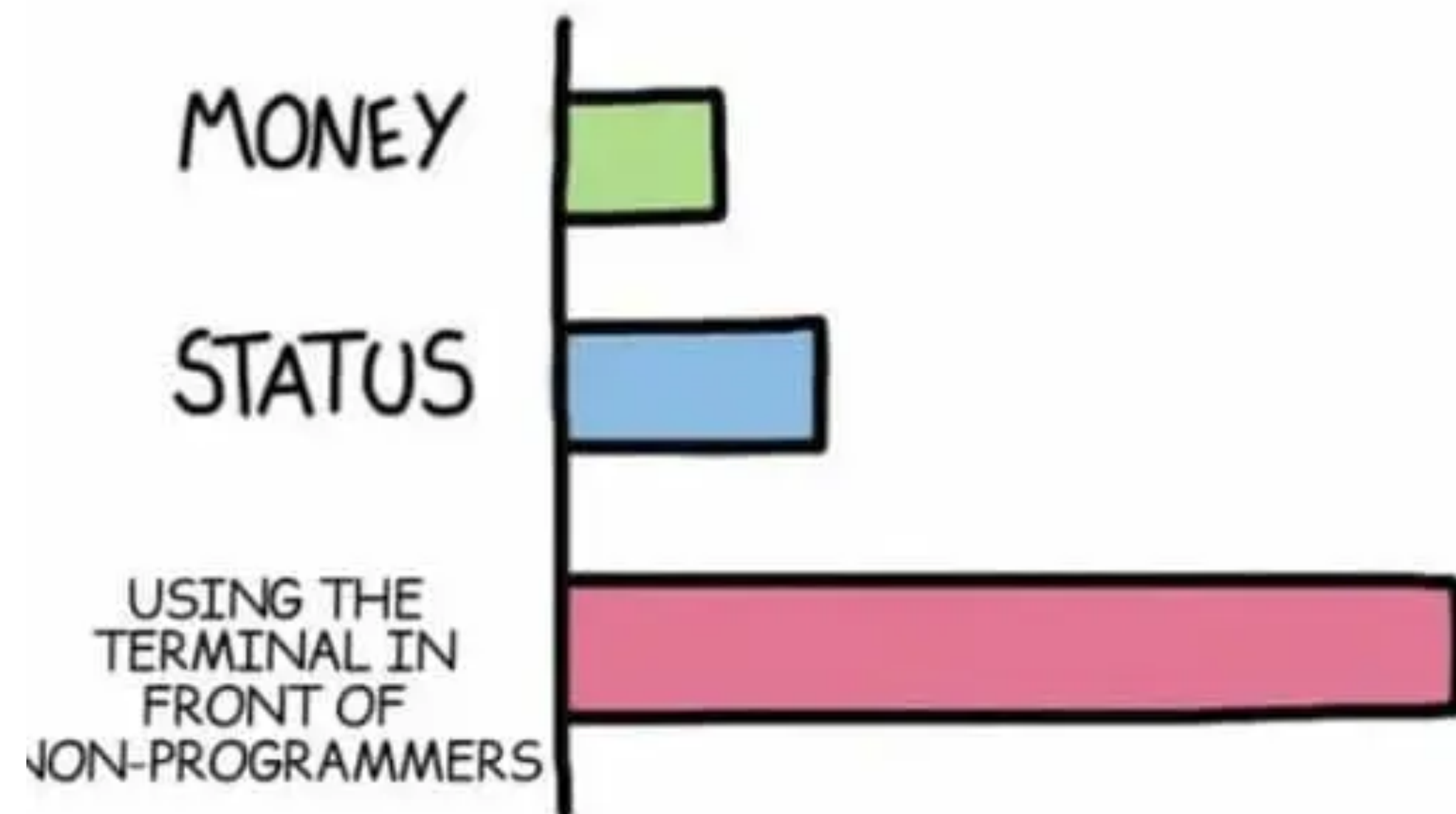
<https://github.com/justgri/ba-programming-2026>

Intro to programming

Where to start?

- Terminal
- VS Code
- Git and GitHub
- HTML and GitHub pages
- Python and Streamlit
- Jupyter Notebook

WHAT GIVES PEOPLE
FEELINGS OF POWER



Homework

Submit self-assessment on OLAT

Homework (~3 hours)

Getting over the first hump

- Watch this Intro to Python and Intro to Git
 - Focus on concepts, don't worry about the syntax (yet)
- Complete software installation (ask for help)
 - Python, VS Code, and Git
- GitHub setup
 - Create an account on <https://github.com/>
 - Clone course repo (how-to - use SSH, verify with git status)
 - Create your first personal repo

Homework (~2 hours)

First contact with the world

- Output (pick your battles - **no AI written code**)
 - Create your first HTML file and host it on GitHub pages
 - Create your first python file and run it in terminal or VS Code
 - Create a Jupyter notebook (.ipynb file) and run it on VS Code
 - Create your first streamlit app and host it on Streamlit community cloud
- Fundamentals
 - Textbook (ch. 1 & ch. 2 & ch. 3)
 - W3 Schools (intro topics up to Python If...Else)